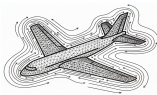


Hybridization (Numerical Solvers)

Numerics-Informed



Closure
Modelling

Fully Coupled



Correction/
Acceleration

NUMERICAL
SOLVER



GIVEN
PHYSICAL
PROBLEM

Model Architecture (Inductive Bias)

Non-Geometric
Data



MLP

Spatial Grid
Data
(e.g. Images)



CNN

Graph/Mesh
Data



GNN

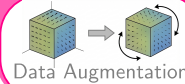
Temporal
Data
(e.g. Time Series)



RNN

Training Procedure (Physical Priors)

Physics-
Aware
(Soft Constraints)



Data Augmentation

Physics-
Guided
(Strong Constraints)



Vector
Potential

Physics-
Informed

$$\text{Loss} = \mathcal{L}_{PDE} + \lambda \mathcal{L}_{BC}$$

Physics-
Embedded



Differentiable
solver